

Demand Forecasting & Inventory Optimization

Jelin Foo

Problem Statement

Many companies struggle with inventory forecasting - inaccurate stock levels caused by legacy practices, obsolete approaches, and unpredictable demand patterns and surges.

Project goal

Identify

sales demand patterns through exploratory analysis

Build

models that predict sales quantity and forecast future demand

Optimize

inventory levels and minimize overstocking/excess stocks

The Path to Demand Prediction

1

Import Data

Load dataset into Jupyter with SQLite and pandas

2

Data Cleaning

Remove missing values, verify datatypes, handle partial months

3

Exploratory Data Analysis (EDA)

Visualize trends, distributions, and feature relationships

4

Regression Models

Linear Regression, Random Forest, XGBoost

5

Time Series Forecasting

ARIMA and SARIMA to forecast future monthly demand

6

Conclusion

Findings, business recommendations, and limitations

Dataset Overview

73,100

rows of daily records

Jan'22 – Jan'24

date coverage (24 complete months)

5

product categories

15

columns

Key fields

Sales Qty Target variable - units sold per product per day

Stock Level Inventory on hand - the strongest predictor

Price, Discount, Competitor Pricing Commercial variables

Weather, Seasonality, Holiday/Promotion External factors

Category, Product, Store, Region Dimensional attributes

Demand Forecast Reference

01

Data Cleaning & EDA

Understanding demand patterns before modeling

Sales Patterns Are Stable

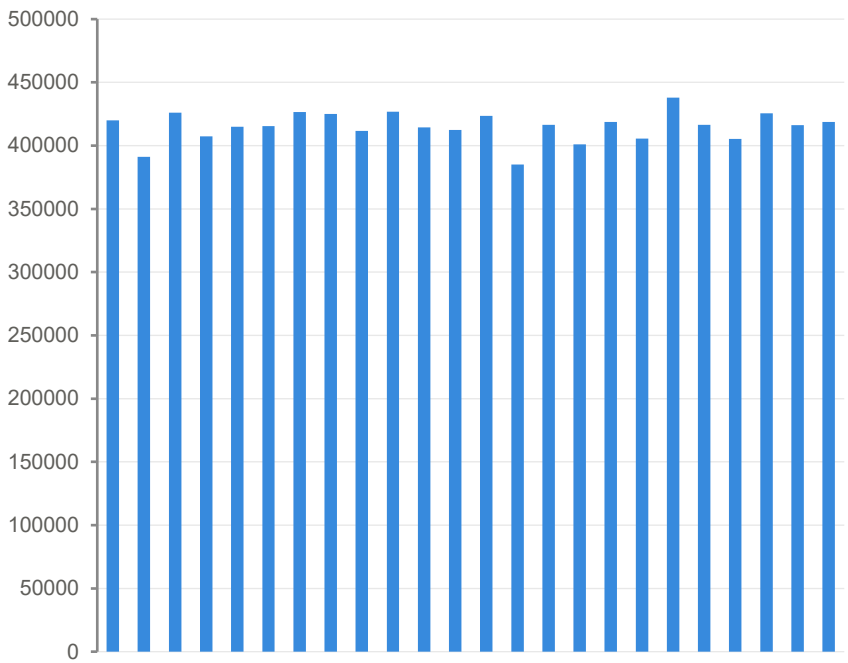
Daily sales Consistent fluctuation within a defined range - no sharp spikes or drastic declines

Monthly sales Stable between 385K and 437K units per month across 2022-2023

Distribution Most daily sales fall between 0-100 units. Monthly sales fall between 300-400K units

Top Sales by Category Furniture - 20.3% of overall sales

Monthly sales (units, Jan 2022 – Dec 2023)



Flat trend - no growth or decline

External Factors – weak indicator

Average daily sales are identical across all external conditions:

Weather

Sunny 138 · Cloudy 136 · Snowy 135 · Rainy 135

3-unit spread

Seasonality

Autumn 137 · Winter 136 · Spring 135 · Summer 135

2-unit spread

Holiday / Promotion

Normal day 136 · Holiday/Promotion 136

0-unit spread

Correlation analysis confirms: Stock Level is the only meaningful predictor (0.59). Price, Discount, Competitor Pricing, Order Qty, and Holiday/Promotion all sit at ~0.00 - behaving as noise for linear models.

02

Regression Models

From linear baselines to gradient boosting

Regression Models

#	Model	Features	Train R ²	Test R ²
1	Linear Regression	Stock Level	0.348	0.348
2	Random Forest	Stock Level	0.354	0.343
3	Linear Regression	Stock Level + Category + Order Qty	0.348	0.348
4	XGBoost	All 16 features	0.507	0.308
5	XGBoost (tuned)	All 16 features	0.356	0.346

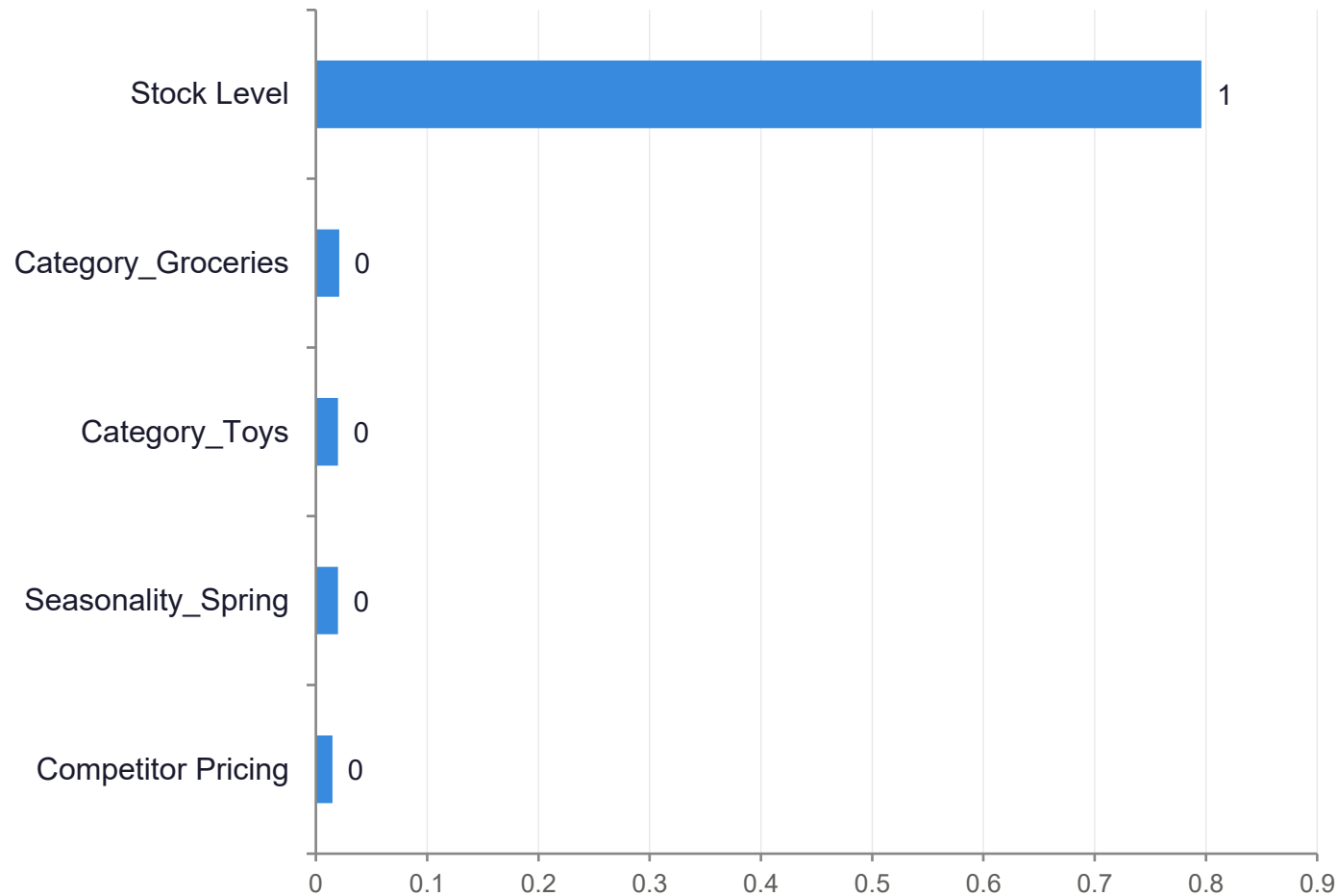
All models plateau at ~34–35% Test R²

The dataset carries weak signal beyond Stock Level - limiting predictive power regardless of model complexity.

What R² = 0.35 means

The model explains ~35% of the variation in sales quantity. The remaining ~65% is driven by factors not present in this dataset.

Stock Level Drives ~80% of Predictions



Validated three ways

1. EDA - flat averages across external factors
2. Correlation - Stock Level 0.59, others ~0.00
3. Model - XGBoost feature importance chart confirms Stock Level as the most dominant factor of the Top 5 important features

Business takeaway

Stock availability is the primary driver of sales in this dataset - the remaining 15 features combined contribute only ~20%.

03

Time Series Forecasting

ARIMA & SARIMA - predicting next month's demand

Forecasting Approach

1

Prepare Aggregate daily sales into 24 complete monthly totals; exclude the 1-day partial month (Jan 2024)

2

Validate ADF test confirms stationarity ($p = 0.0000$) - no differencing required ($d = 0$)

3

Split Train on 21 months (Jan 2022 - Sep 2023); last 3 months (Oct - Dec 2023) as unseen test data

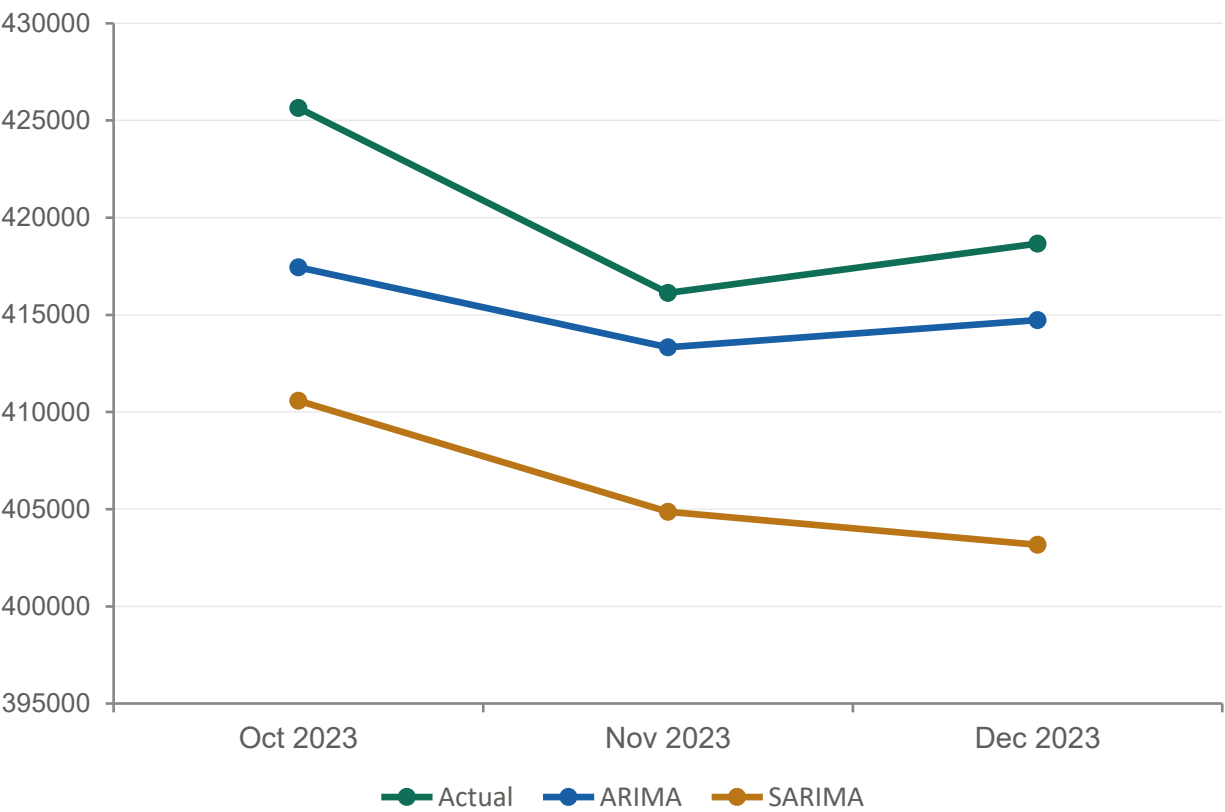
4

Select Auto ARIMA finds optimal parameters - ARIMA(1,0,0) and SARIMA(1,0,0)(0,0,1,12)

5

Evaluate Compare forecasts to actual sales using MAE, RMSE, and MAPE

ARIMA Wins: 1.2% Forecast Error



Metric	ARIMA(1,0,0)	SARIMA(1,0,0)(0,0,1,12)
MAE	4,978 units	13,941 units
RMSE	5,494 units	14,071 units
MAPE	1.2%	3.3%

Why SARIMA underperformed

With only 21 training months (< 2 seasonal cycles), the seasonal component learned an unreliable pattern from Oct-Dec 2022 and applied a systematic downward bias to 2023 forecasts. Simpler models proved better.

Key Findings

1

Stock Level dominates

Validated three ways – correlation matrix at 0.59, flat EDA averages and 79.6% XGBoost feature importance.

2

Regression ceiling at ~35%

All five models plateau at 34-35% Test R^2 - the dataset carries weak signal beyond stock availability.

3

ARIMA is the best forecaster

1.2% MAPE on 3-month-ahead forecasts, outperforming SARIMA (3.3%) which lacked seasonal history.

4

Simpler models won consistently

In both regression and time series, added complexity failed to beat simpler baselines on this dataset.

Business Recommendations

1

Adopt ARIMA monthly forecasting

Average error of ~5,000 units.

2

Hold a safety stock buffer

Covers the observed forecast error range while minimising overstocking.

3

Adding in Supplier Lead time

Prevents out of stock situation with predictable range of lead time

4

Re-train the model monthly

Accumulate actual sales qty each month to improve future forecast accuracy

Limitations & Future Work

Limitations

Synthetic dataset

Near-zero decomposition residuals and artificially balanced categories inflate forecast accuracy - real-world MAPE typically runs 10-30%.

Limited seasonal history

24 months is insufficient for reliable SARIMA - 36 to 60 months recommended for seasonal modeling.

Weak feature signal

~65% of sales variance remains unexplained by available features.

Future work

Apply to real-world data

Validate the full framework on genuine transaction history with richer demand signals.

SKU-level forecasting

Top-down allocation of the monthly forecast to individual products using historical proportions.

Data collection

Collect more data (36-60 months) to view demand patterns and trends over long periods and add in lead time features